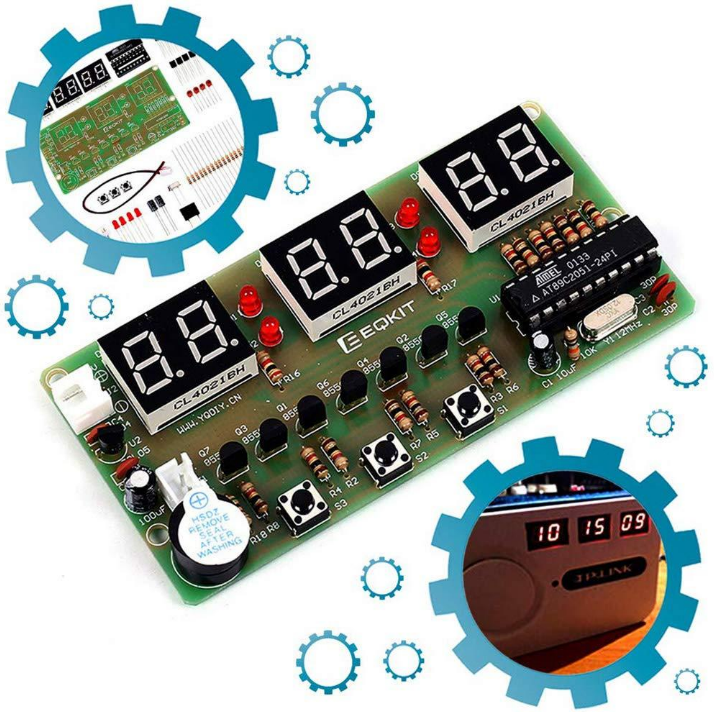
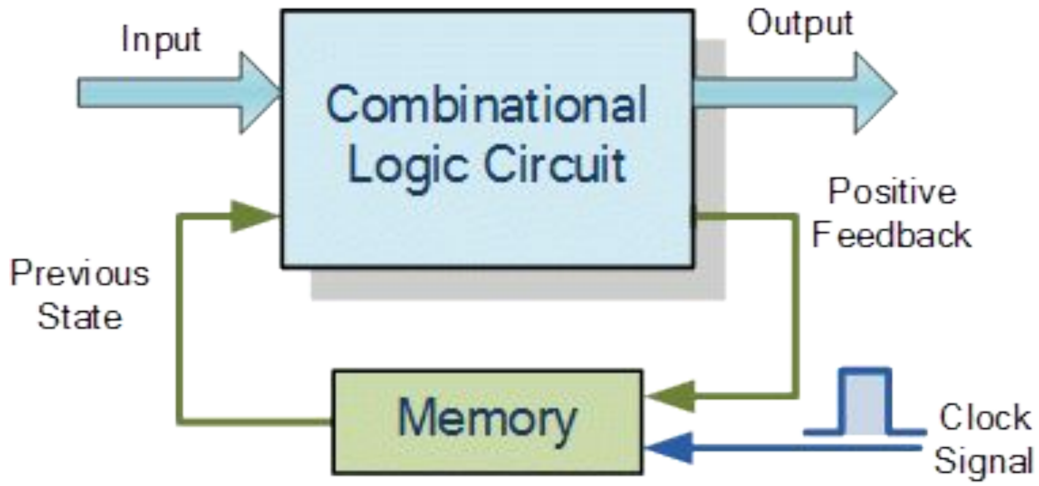


Dr. Hiran Ekanayake

SEQUENTIAL LOGIC CIRCUITS

Lesson Outline

- Sequential logic circuits introduction
- S-R flip-flop / latch
- Timing diagrams
- J-K flip-flop / latch
- D flip-flop / latch

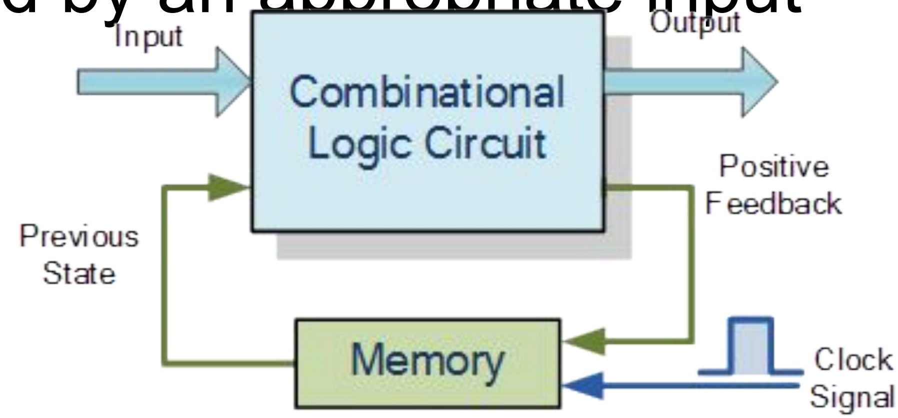


SEQUENTIAL LOGIC CIRCUITS

https://www.electronics-tutorials.ws/sequential/seq_1.html

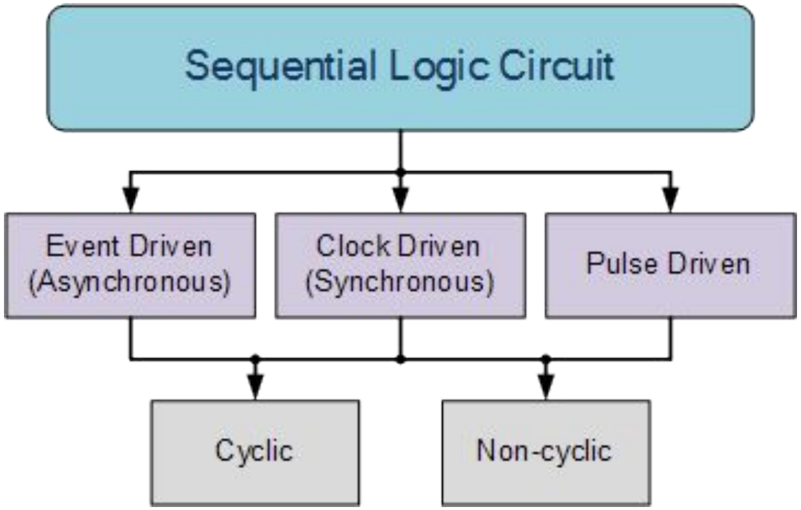
What is a sequential logic circuit?

- It is a combinational logic circuit with a memory
- The output state of a sequential logic circuit depends on the "present input", the "past input" and/or the "past output"
- It is sometimes called a two state or bistable device since it will remain "latched" (locked) indefinitely in its current state or condition until it gets triggered by an appropriate input



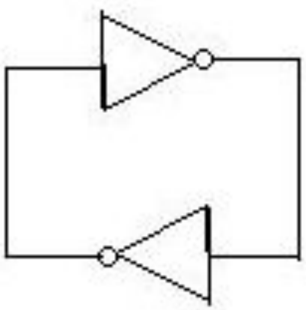
What are different types of sequential logic circuits?

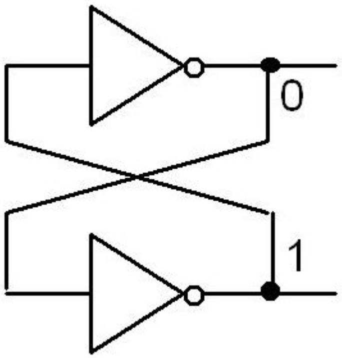
- Event driven
  - Asynchronous circuits that change state immediately when enabled
- Clock driven
  - Synchronous circuits that are synchronized to a specific clock signal
- Pulse driven
  - Which is a combination of the two that responds to triggering pulses

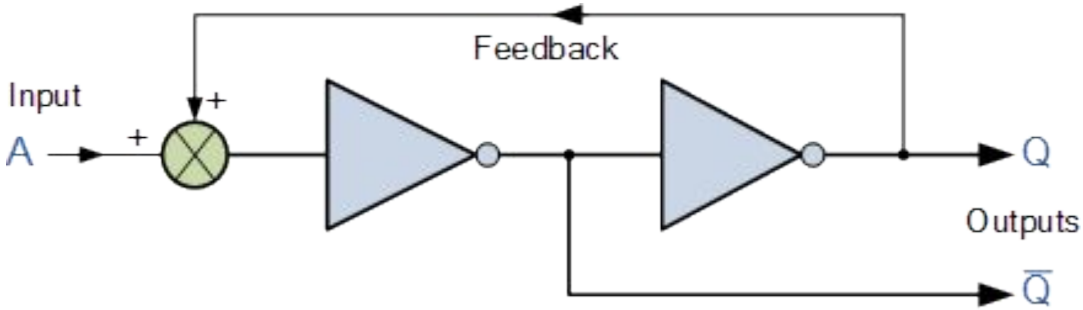


Sequential Feedback Loop

- How do you construct a memo cell using two inverters?
- What are the drawbacks of the above arrangement?
- How do you overcome those drawbacks?

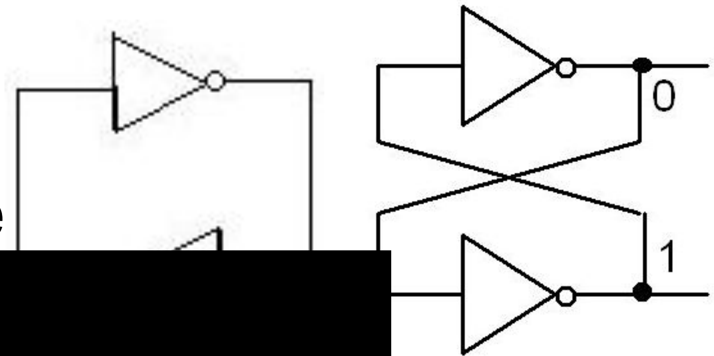


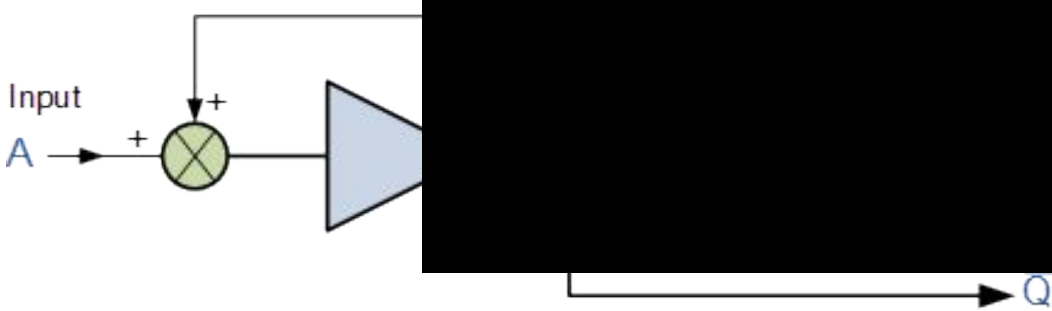




Sequential Feedback Loop

- How do you construct a memo cell using two inverters?
- What are the drawbacks of the above arra [redacted]
- How do yo [redacted] drawbacks [redacted]





Sequential Feedback Loop

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- What are the drawbacks of the above arrangement?
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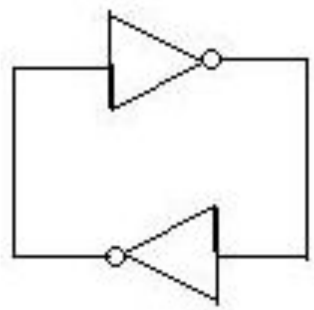
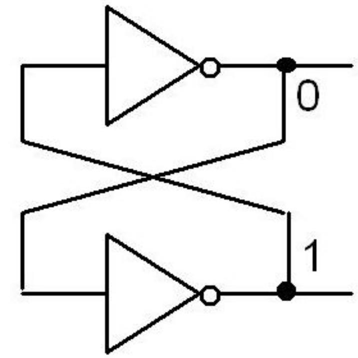
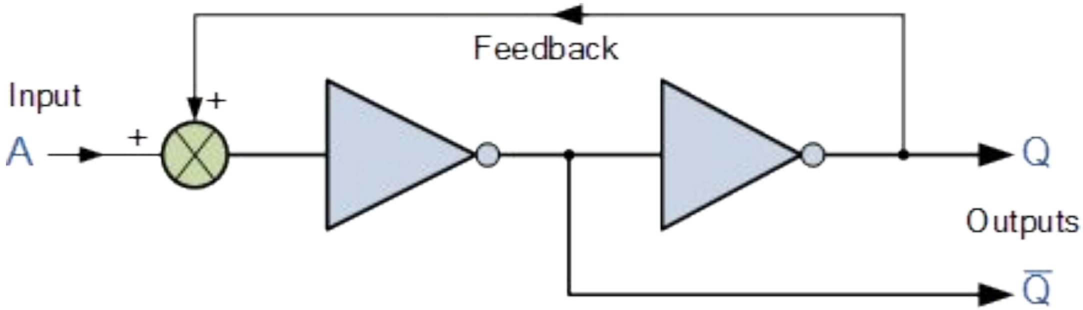
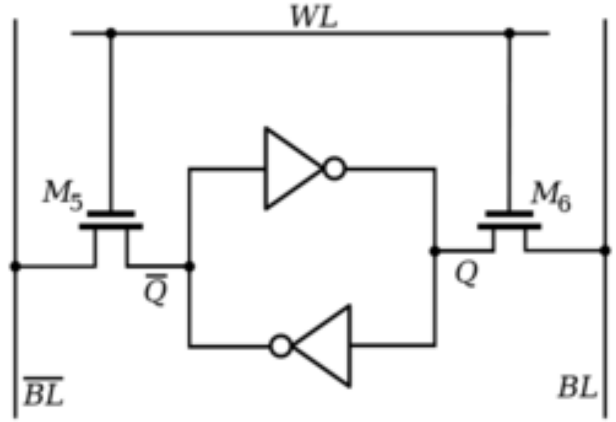


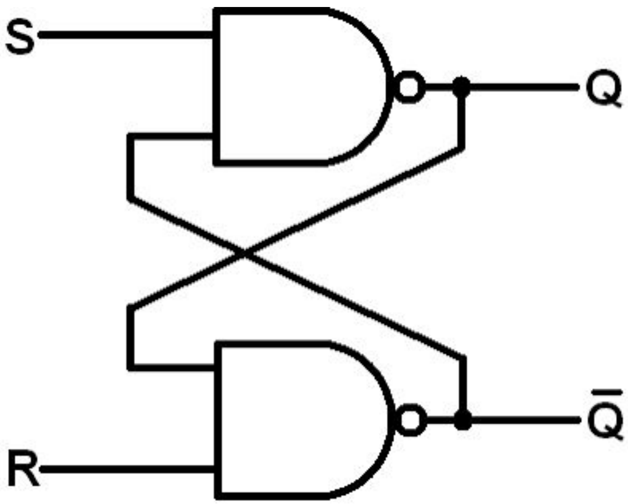


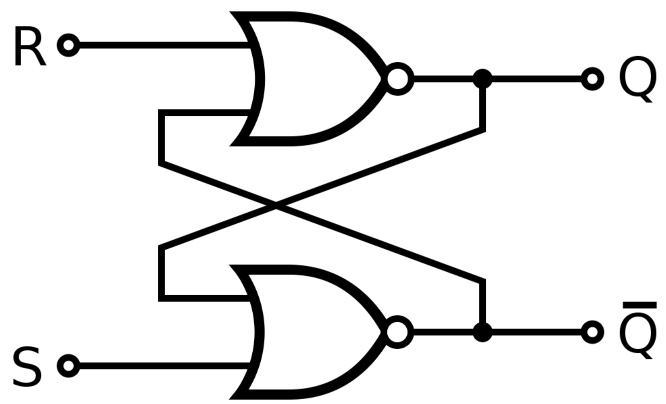




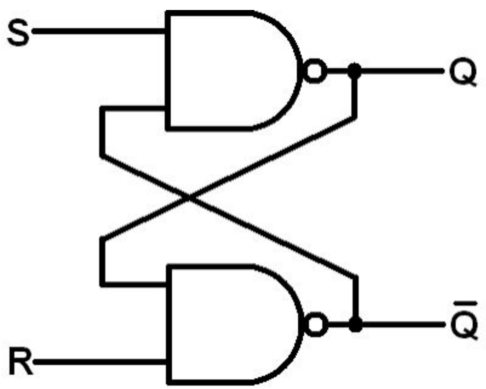
Set-Reset (S-R) Flip-Flop / Latch

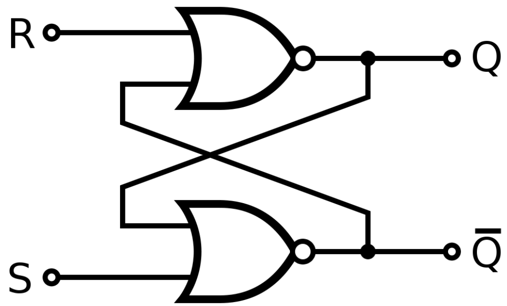
- A one-bit memory bistable device that has two inputs: “SET” the output to “1” and “RESET” the output to “0”
  - Note: flipping=SET, flopping=RESET





Set-Reset (S-R) Flip-Flop / Latch



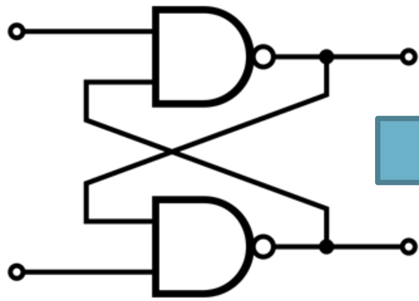


State	S	R	Q		Description
Set	1	0	0	1	Set $\overline{Q}$ » 1
	1	1	0	1	no change
Reset	0	1	1	0	Reset $\overline{Q}$ » 0
	1	1	1	0	no change
Invalid	0	0	1	1	Invalid Condition

S	R	Q	$\overline{Q}$
0	0	No change	
0	1	1	0
1	0	0	1
1	1	X	X
		(Invalid)	

Set-Reset (S-R) Flip-Flop / Latch

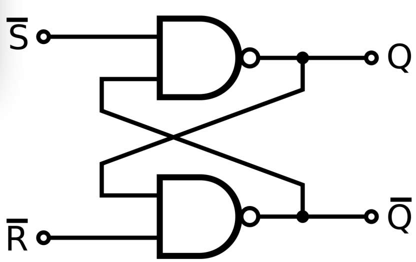
00		01
10		11



```
Truth Table...
0 0 => 1 1
0 1 => 1 0
1 0 => 0 1
1 1 => 0 1
0 0 => 1 1
1 1 => 0 1
1 0 => 0 1
0 1 => 1 0
1 1 => 1 0
0 1 => 1 0
0 0 => 1 1
1 0 => 0 1
0 0 => 1 1
Truth Table...
0 0 => 1 1
0 1 => 1 0
1 0 => 0 1
1 1 => 0 1
0 0 => 1 1
1 1 => 0 1
```

I1	I2	O1	O2
0	0	1	1
0	1	1	0
1	0	0	1
1	1	X	X

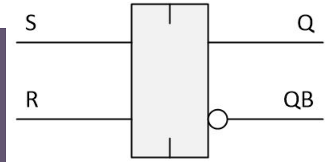
S	R	Q	$\bar{Q}$
0	0	No change	
0	1	0	1
1	0	1	0
1	1	X	X
		(Invalid)	



$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$
1	1	1	1
1	0	1	0
0	1	0	1
0	0	X	X

Latch vs. Flip-Flop

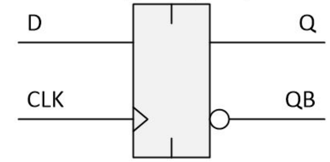
SR Latch



R	S	$Q_{(new)}$	$QB_{(new)}$
0	0	$Q_{(old)}$	$QB_{(old)}$
0	1	1	0
1	0	0	1
1	1	0*	0*

* Unstable configuration

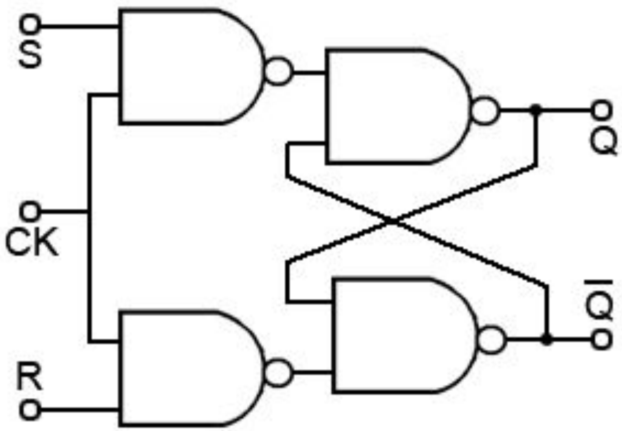
D-Type Flip-Flop

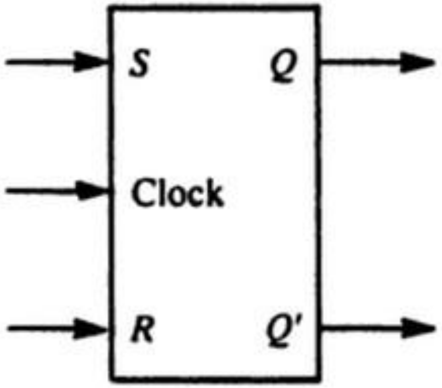


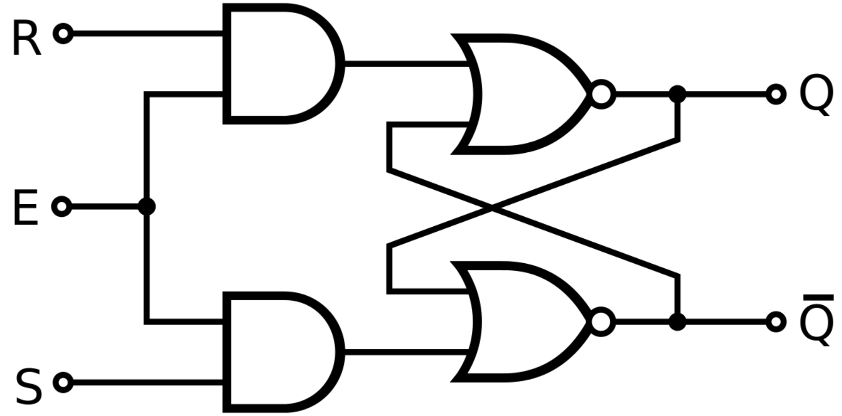
CLK	D	$Q_{(new)}$	$QB_{(new)}$
↑	0	0	1
↑	1	1	0
0	X	$Q_{(old)}$	$QB_{(old)}$
1	X	$Q_{(old)}$	$QB_{(old)}$
↓	X	$Q_{(old)}$	$QB_{(old)}$

Feature	Latch	Flip-Flop
Triggering	Level-sensitive	Edge-sensitive
Clock Dependency	Not required	Required
Timing	Continuous updates	Updates on clock edge
Use Case	Asynchronous circuits	Synchronous circuits
Complexity	Simpler	More complex

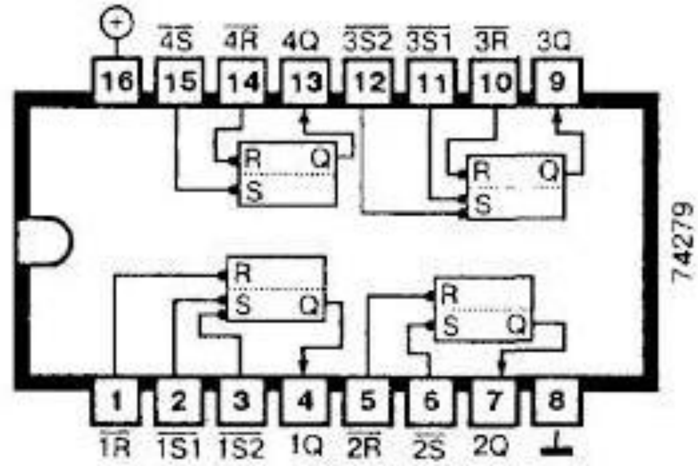
Gated / Clocked Set-Reset (S-R) Flip-Flop / Latch



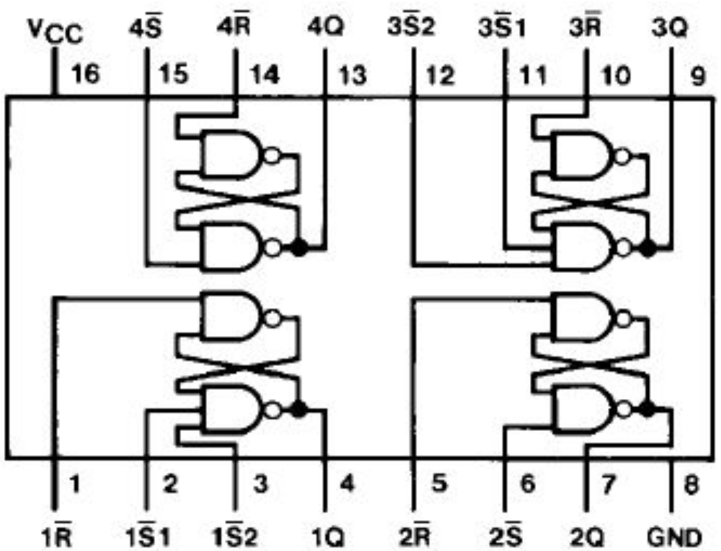




74LS279 Quad S-R Latch



Connection Diagram



Function Table

Inputs		Output
$\overline{S}$ (Note 1)	$\overline{R}$	Q
L	L	H (Note 2)
L	H	H
H	L	L
H	H	Q_0

H = HIGH Level

L = LOW Level

Q_0 = The Level of Q before the indicated input conditions were established.

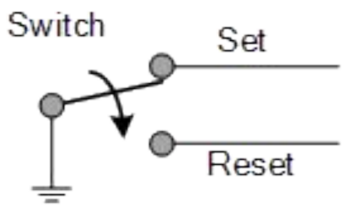
Note 1: For latches with double $\overline{S}$ inputs:

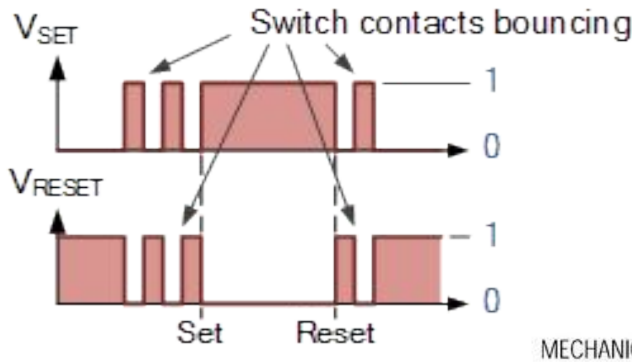
H = both $\overline{S}$ inputs HIGH

L = one or both $\overline{S}$ inputs LOW

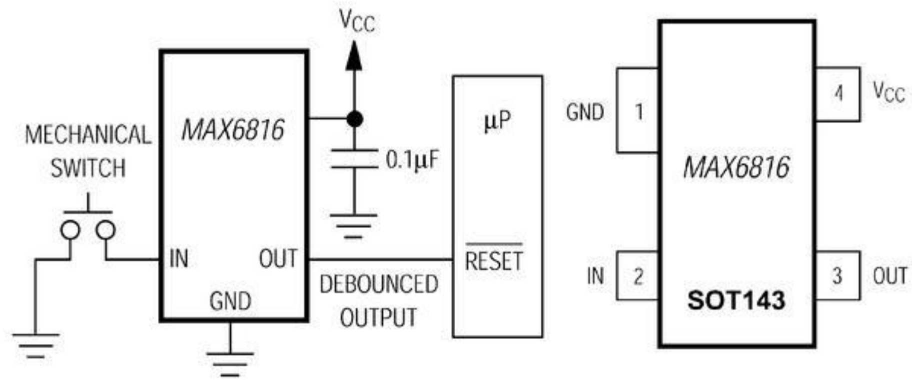
Note 2: This output level is pseudo stable; that is, it may not persist when the $\overline{S}$ and $\overline{R}$ inputs return to their inactive (HIGH) level.

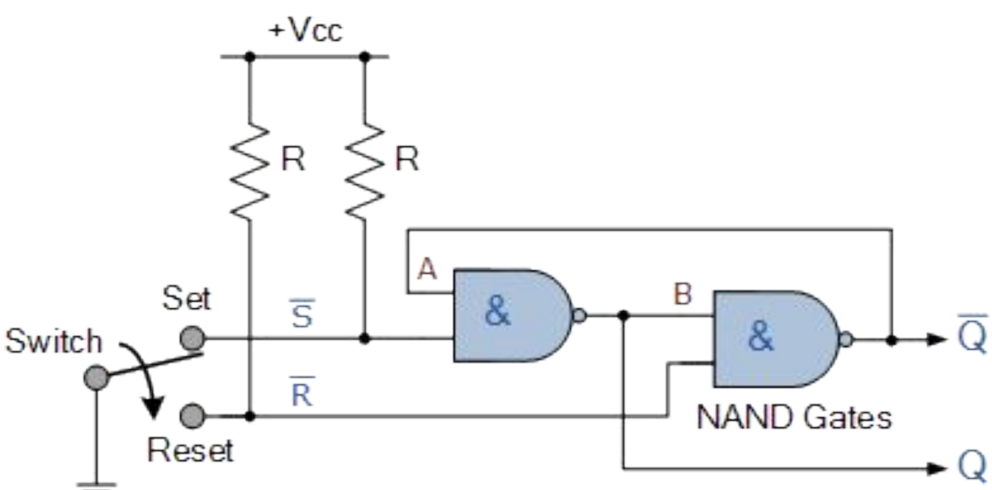
Application: Switch Debounce

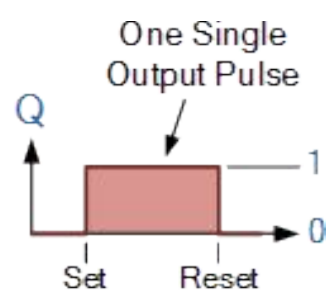




Multiple signals (false inputs)

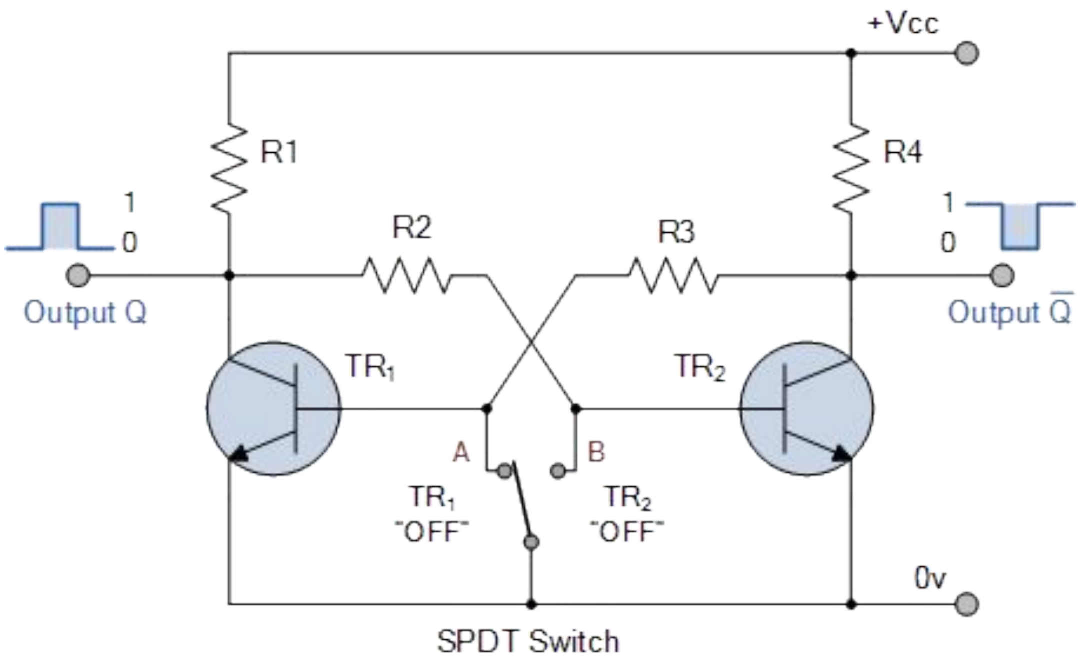




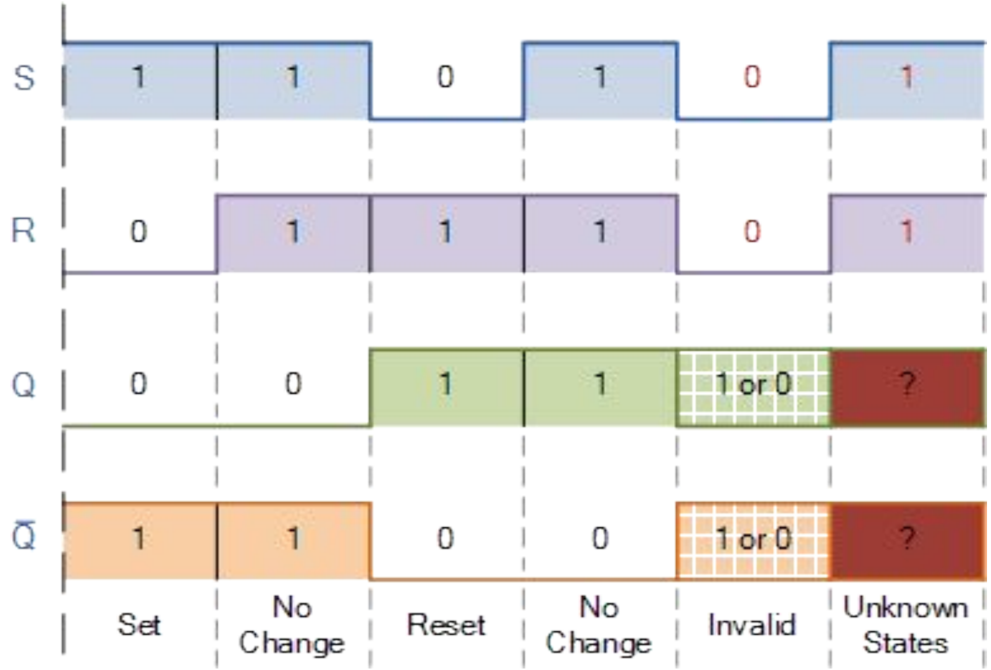


Transistor-based SR Flop-Flop

- How do you construct a transistor-based SR flip-flop?



In the circuit, two cross-coupled transistors are operating as “ON-OFF” transistor switches. In each of the two states, one of the transistors is cut-off while the other transistor is in saturation, this means that the bistable circuit can remain indefinitely in either stable state.



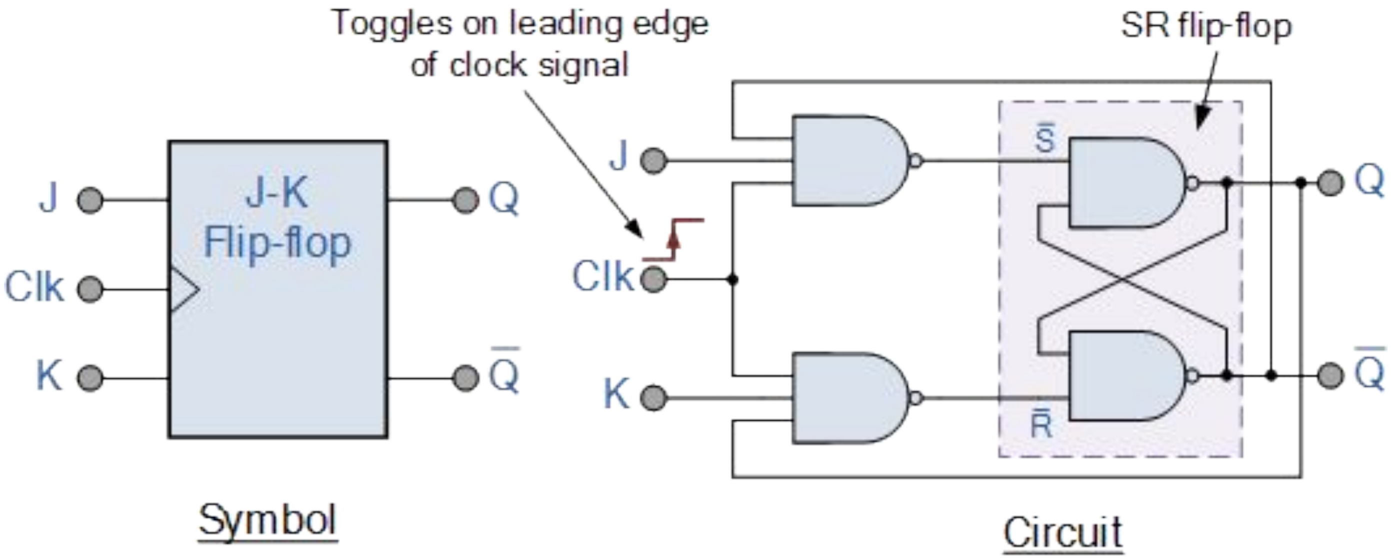
TIMING DIAGRAMS

SR Flip-Flop Timing Diagram

```
Time ————————————————————>

CLK : __|‾|__|‾|__|‾|__|‾|__

S   : _______|‾‾‾|__________

R   : ______________|‾‾|__

Q   : ____0____1____1____0
```

Rising Edge	S	R	Q	Action
↑1	0	0	0	Hold
↑2	1	0	1	Set
↑3	0	0	1	Hold
↑4	0	1	0	Reset

- Only the **rising edge (↑)** of the clock causes the flip-flop to sample the inputs.
- Output changes **only at the active clock edge**.
- Between clock edges, changes in **S** or **R** do not affect **Q**.
- The SR flip-flop stores one bit of memory.

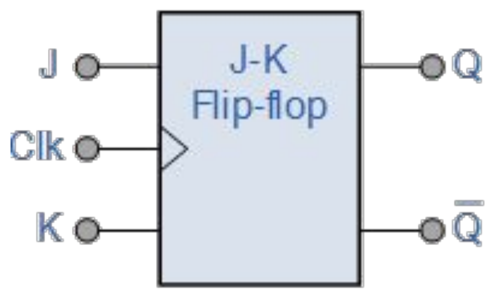


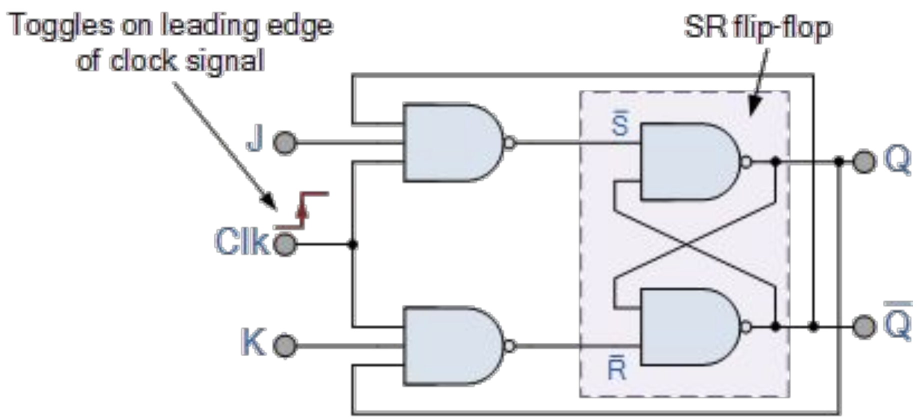
JK (JACK KILBY) FLIP-FLOP

JK Flip-Flop

- Why JK flip-flop?
  - SR flip-flop has several drawbacks: invalid input conditions (S=R=0/1) should be avoided, there will be incorrect outputs if the SR inputs change the states while EN,CLK=1, etc.
- Called the universal flip-flop
- No invalid/forbidden input states
  - When J=K=0□No Change, J=K=1□Toggle

JK Flip-Flop



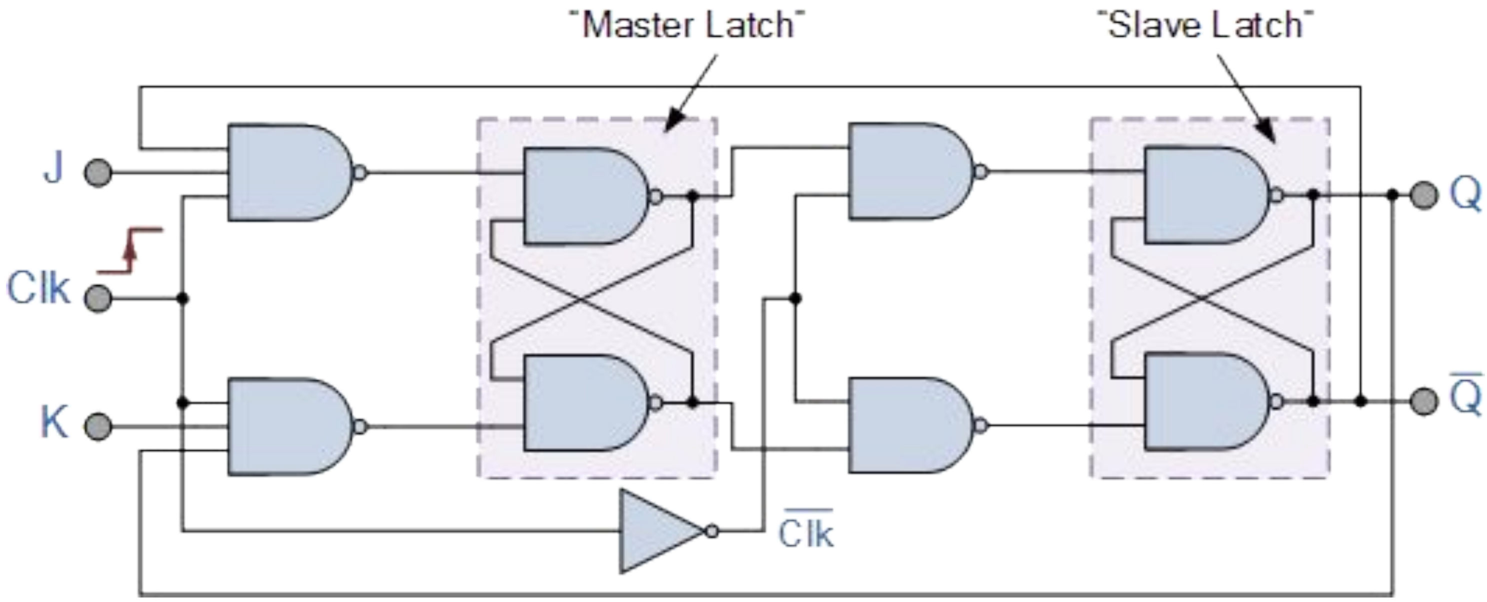


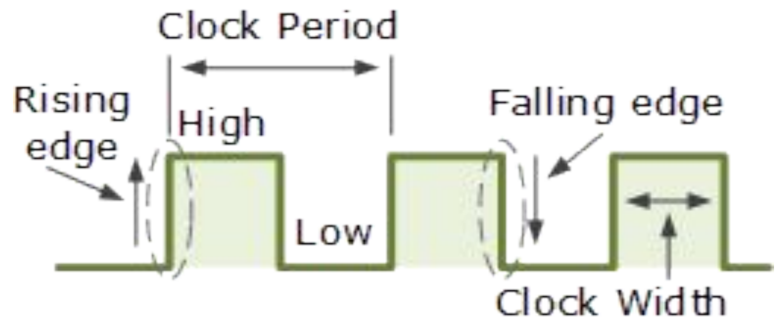
	Clock	Input		Output		Description
	Clk	J	K	Q	$\overline{Q}$	
same as for the SR Latch	X	0	0	1	0	Memory no change
	X	0	0	0	1	
	¯↓_	0	1	1	0	Reset Q » 0
	X	0	1	0	1	
	¯↓_	1	0	0	1	Set Q » 1
	X	1	0	1	0	
toggle action	¯↓_	1	1	0	1	Toggle
	¯↓_	1	1	1	0	

JK Flip-Flop: Race Condition

- What is race condition in sequential logic circuits?
  - The output state changes several times while CLK=1 (when it acts as EN)
- How do you avoid the race condition?
  - Make the CLK pulse very short
  - Make use of the rising (leading) and/or falling edges of the CLK pulse
- What is the difference between CLK and EN?

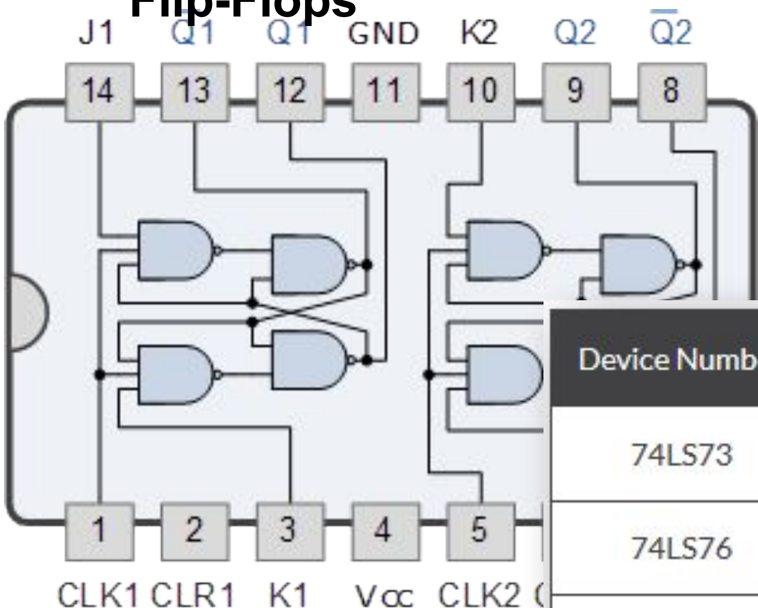
Master-Slave JK Flip-Flop



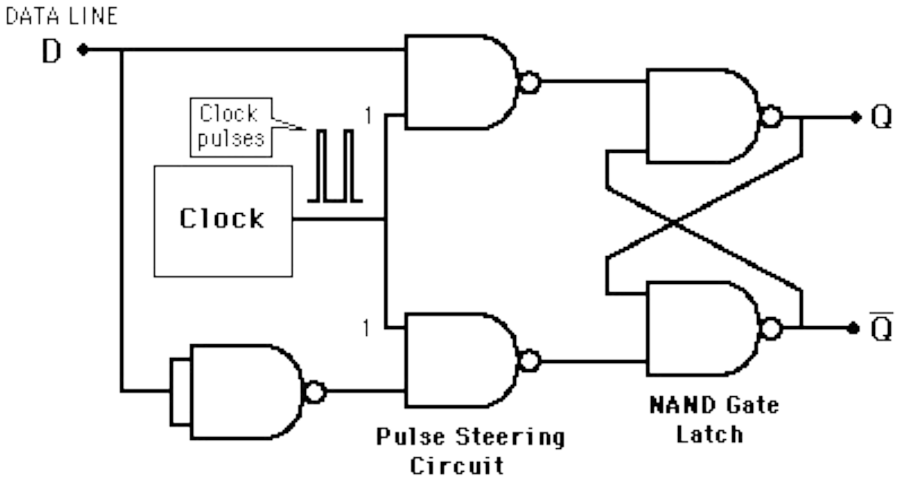


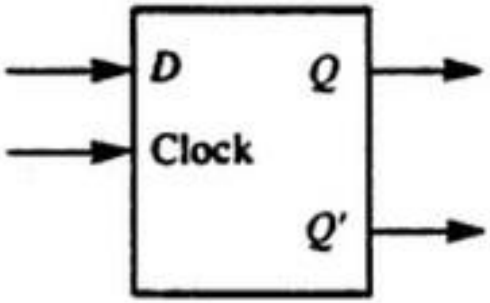
JK Flip-Flop ICs

74LS73 Dual JK Flip-Flops



Device Number	Subfamily	Device Description
74LS73	LS TTL	Dual JK-type Flip Flops with Clear
74LS76	LS TTL	Dual JK-type Flip Flops with Preset and Clear
74LS107	LS TTL	Dual JK-type Flip Flops with Clear
4027B	Standard CMOS	Dual JK-type Flip Flop



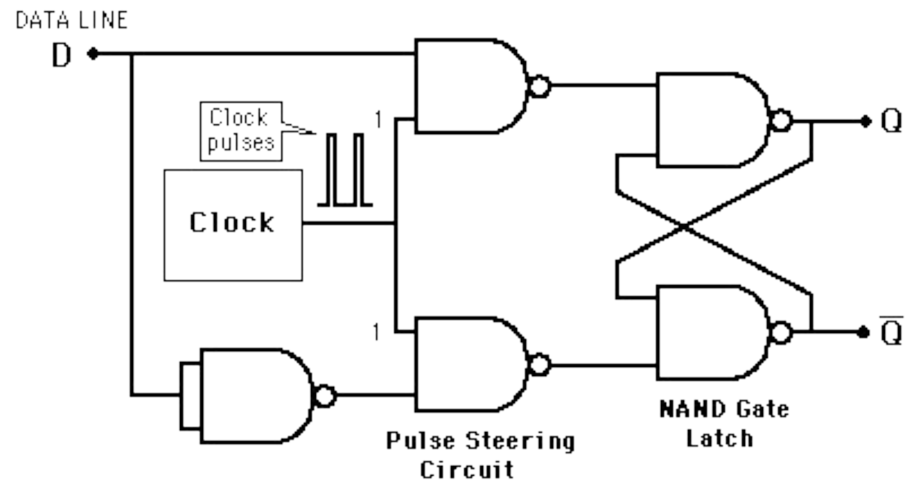


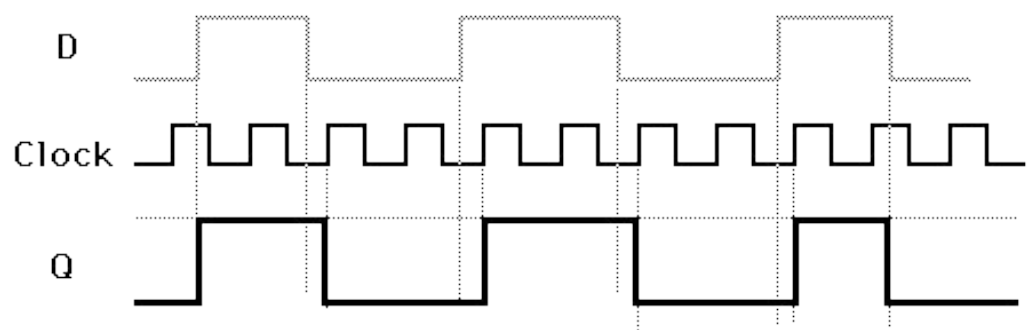
DATA (D) FLIP-FLOP

D Flip-Flop

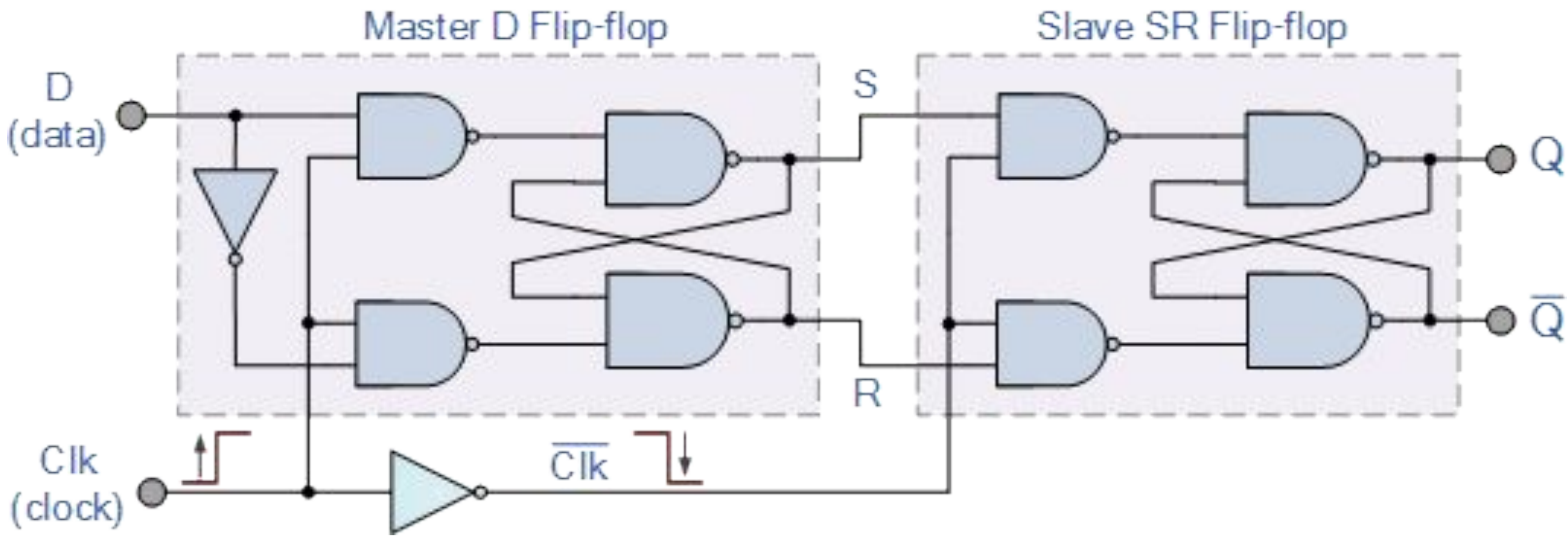
- Why D-type flip-flop?
  - It avoids the S=R or J=K condition

Clk	D	Q		Description
↓ » 0	X	Q	$\overline{Q}$	Memory no change
↑ » 1	0	0	1	Reset Q » 0
↑ » 1	1	1	0	Set Q » 1



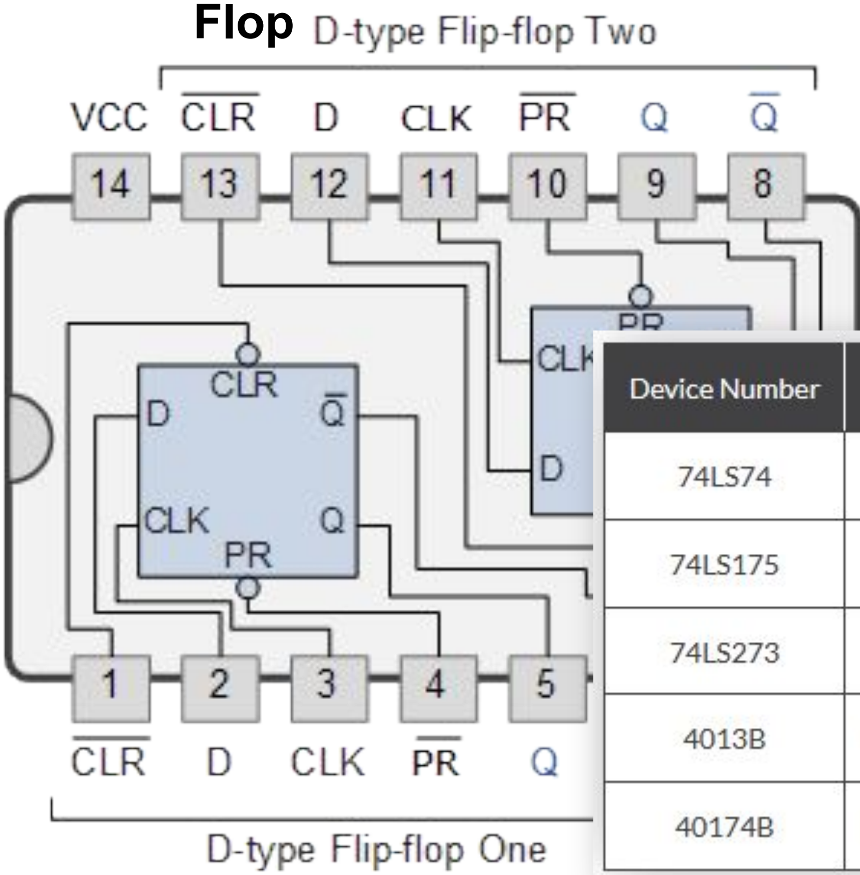


Master-Slave D Flip-Flop



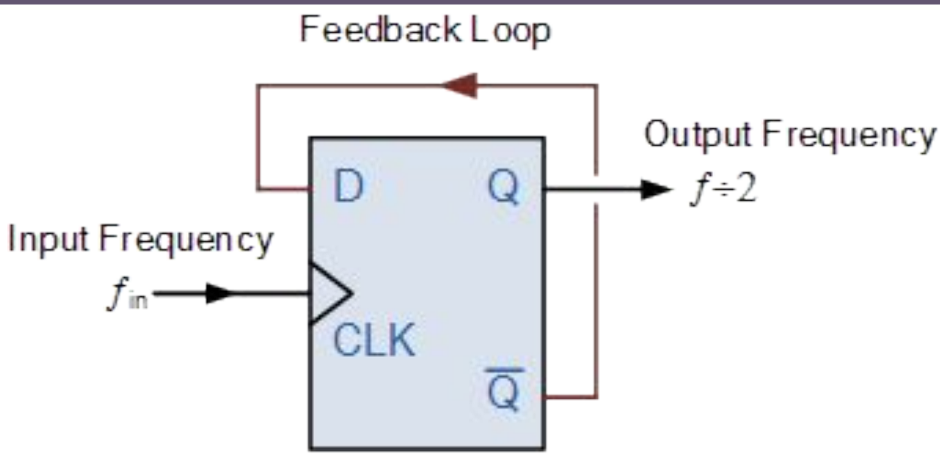
D Flip-Flop ICs

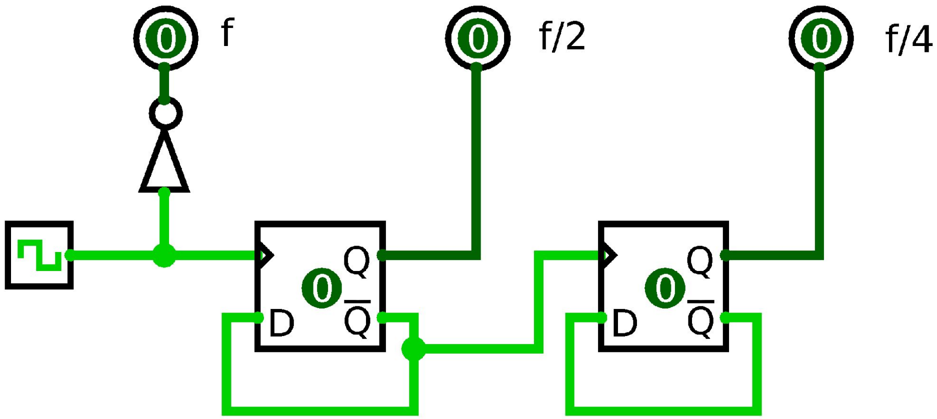
74LS74 Dual D-type Flip Flop

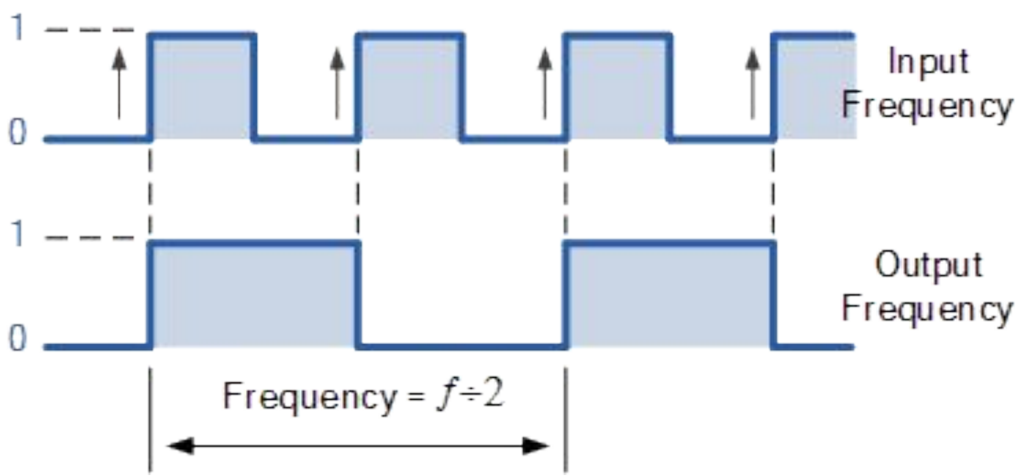


Device Number	Subfamily	Device Description
74LS74	LS TTL	Dual D-type Flip Flops with Preset and Clear
74LS175	LS TTL	Quad D-type Flip Flops with Clear
74LS273	LS TTL	Octal D-type Flip Flops with Clear
4013B	Standard CMOS	Dual type D Flip Flop
40174B	Standard CMOS	Hex D-type Flip Flop with Master Reset

D Flip-Flop Application: Divide-By-2







D Flip-Flop Application: Data Latch